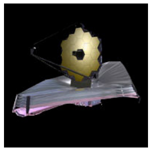

10752 - A Deep Spectral Imaging View into Two CMZ Massive Protostars

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
Observation Folder				
	1	NIRSPEC-IFU	NIRSpec IFU Spectroscopy	(1) G359.44a
	2	NIRSPEC-IFU	NIRSpec IFU Spectroscopy	(3) G359.44b
	3	MIRI-IFU	MIRI Medium Resolution Spectroscopy	(4) G359.44a-MIRI
	4	MIRI-IFU	MIRI Medium Resolution Spectroscopy	(5) G359.44b-MIRI
	5	MIRI-IFU-BKG	MIRI Medium Resolution Spectroscopy	(2) SGR-C-BKG

ABSTRACT

We propose JWST-NIRSpec and -MIRI-MRS IFU observations of the Central Molecular Zone (CMZ) molecular cloud Sagittarius C, in particular the massive protostars G359.44a and G359.44b and their immediate surrounding region, to characterize their outflows, warm gas envelopes, nearby

sources, and chemical composition, and probe the unique circumstances of massive star formation in this extreme environment. JWST-NIRCam imaging and ALMA have revealed outflows in the vicinity of the protostars in ionized and molecular hydrogen, as well as extended molecular jets from each source. These data have provided much information about the wider environment surrounding G359.44a and G359.44b, however they lack specific characteristic detail about the massive protostars, their outflows, and their close neighbors. The proposed JWST observations have three goals: 1) characterize the spatial, kinematic, and dynamic properties of the protostellar jets powered by G359.44a and G359.44b, including the temperature, mass, luminosity, and, mass loss rate. We will also characterize the neighboring YSOs around each protostar, with ramifications for theories of massive star formation. 2) Probe the warm and hot gas content, constraining the effects of outflow feedback very close to each protostar; 3) Detect and classify ice and silicate features in the disk and infall envelope to assess hot core chemistry in the extreme environment of the CMZ.

OBSERVING DESCRIPTION

We will use NIRSpec and MIRI-MRS in IFU mode to observe the heart of the massive star-forming region Sgr C, the massive protostars G359.44a and G359.44b. We will have continuous coverage both in spectra and imaging from 2.87 to 28 microns.

We will observe with NIRSpec the protostars and their close environments with a 4-dither pattern in a single pointing without any PA constraint. To meet our goals we will observe with NRSIRS2 Readout Pattern, 6 Groups per integration, and 2 Integrations per exposure, for a total integration time of 3618s. For all dithers and filters, we will obtain leakage measurements (with half of the science time) to account for the complex environment and flux leakage.

We will also observe each protostar with MIRI-MRS with all three sub-bands of the four channels. All observations have a 4-dither pattern in a single pointing without PA constraint. We set the observations to FASTR1 Readout Pattern, 36 Groups per integration, and 1 Integration per exposure, for a total integration time of 399.606s. We have also set a background measurement close to the science targets, but far from their nebulosity, with identical configurations and integration times, as well as simultaneous MIRI imaging for accurate astrometry.

All five visits have the special requirement of being non-interruptible.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	G359.44a	RA: 17 44 40.1505 (266.1672937d) Dec: -29 28 12.91 (-29.47025d) Equinox: J2000	Epoch of Position: 2000.0	
Comments: Category=Star Description=[Ejecta, Young stellar objects] Extended=YES				
(2)	SGR-C-BKG	RA: 17 44 43.0000 (266.1791667d) Dec: -29 28 3.00 (-29.46750d) Equinox: J2000		
Comments: Background target for MIRI observations. The exact position will be selected in close conjunction with the STScI team in phase II. Category=Star Description=[Ejecta, Young stellar objects] Extended=YES				
(3)	G359.44b	RA: 17 44 40.5775 (266.1690729d) Dec: -29 28 15.98 (-29.47111d) Equinox: J2000	Epoch of Position: 2000.0	
Comments: Category=Star Description=[Ejecta, Protostars, Young stellar objects] Extended=YES				
(4)	G359.44a-MIRI	RA: 17 44 40.1505 (266.1672937d) Dec: -29 28 12.91 (-29.47025d) Equinox: J2000	Epoch of Position: 2000.0	
Comments: Category=Star Description=[Ejecta, Young stellar objects] Extended=YES				
(5)	G359.44b-MIRI	RA: 17 44 40.5775 (266.1690729d) Dec: -29 28 15.98 (-29.47111d) Equinox: J2000	Epoch of Position: 2000.0	
Comments: Category=Star Description=[Ejecta, Protostars, Young stellar objects] Extended=YES				

Observation

Proposal 10752, Observation 1: NIRSPEC-IFU

Tue Jun 02 18:00:22 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	G359.44a	RA: 17 44 40.1505 (266.1672937d) Dec: -29 28 12.91 (-29.47025d) Equinox: J2000	Epoch of Position: 2000.0	

Comments:
Category=Star
Description=[Ejecta, Young stellar objects]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	NRSIRS2RAPID	11	5	false	true	NONE	4	20	3501.334	174788
2	G395M/F290LP	NRSIRS2RAPID	11	3	true	true	NONE	4	12	2100.8	

Special Requirements

Group Observations 1, 2, 3, 4, 5, Non-interruptible

Observation

Proposal 10752, Observation 2: NIRSPEC-IFU

Tue Jun 02 18:00:22 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	G359.44b	RA: 17 44 40.5775 (266.1690729d) Dec: -29 28 15.98 (-29.47111d) Equinox: J2000	Epoch of Position: 2000.0	

Comments:
Category=Star
Description=[Ejecta, Protostars, Young stellar objects]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	NRSIRS2RAPID	11	5	false	true	NONE	4	20	3501.334	174788
2	G395M/F290LP	NRSIRS2RAPID	11	3	true	true	NONE	4	12	2100.8	

Special Requirements

Group Observations 1, 2, 3, 4, 5, Non-interruptible

Observation

Proposal 10752, Observation 3: MIRI-IFU

Tue Jun 02 18:00:22 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[MIRI-IFU-BKG (Obs 5)]

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	G359.44a-MIRI	RA: 17 44 40.1505 (266.1672937d) Dec: -29 28 12.91 (-29.47025d) Equinox: J2000	Epoch of Position: 2000.0	

Comments:
Category=Star
Description=[Ejecta, Young stellar objects]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F560W	FASTR1	30	1	1	Dither 1	4	4	333.005	
1	SHORT(A)	MRSLONG		FASTR1	36	1	1	Dither 1	4	4	399.606	174788
1	SHORT(A)	MRSSHORT		FASTR1	36	1	1	Dither 1	4	4	399.606	
2		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	
2	MEDIUM(B)	MRSLONG		FASTR1	36	1	1	Dither 1	4	4	399.606	
2	MEDIUM(B)	MRSSHORT		FASTR1	36	1	1	Dither 1	4	4	399.606	
3		IMAGER	F1000W	FASTR1	30	1	1	Dither 1	4	4	333.005	
3	LONG(C)	MRSLONG		FASTR1	36	1	1	Dither 1	4	4	399.606	
3	LONG(C)	MRSSHORT		FASTR1	36	1	1	Dither 1	4	4	399.606	

Special Requirements	
	Group Observations 1, 2, 3, 4, 5, Non-interruptible

Observation

Proposal 10752, Observation 4: MIRI-IFU

Tue Jun 02 18:00:22 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[MIRI-IFU-BKG (Obs 5)]

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	G359.44b-MIRI	RA: 17 44 40.5775 (266.1690729d) Dec: -29 28 15.98 (-29.47111d) Equinox: J2000	Epoch of Position: 2000.0	

Comments:
Category=Star
Description=[Ejecta, Protostars, Young stellar objects]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F560W	FASTR1	30	1	1	Dither 1	4	4	333.005	
1	SHORT(A)	MRSLONG		FASTR1	36	1	1	Dither 1	4	4	399.606	174788
1	SHORT(A)	MRSSHORT		FASTR1	36	1	1	Dither 1	4	4	399.606	
2		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	
2	MEDIUM(B)	MRSLONG		FASTR1	36	1	1	Dither 1	4	4	399.606	
2	MEDIUM(B)	MRSSHORT		FASTR1	36	1	1	Dither 1	4	4	399.606	
3		IMAGER	F1000W	FASTR1	30	1	1	Dither 1	4	4	333.005	
3	LONG(C)	MRSLONG		FASTR1	36	1	1	Dither 1	4	4	399.606	
3	LONG(C)	MRSSHORT		FASTR1	36	1	1	Dither 1	4	4	399.606	

Special Requirements	Group Observations 1, 2, 3, 4, 5, Non-interruptible

Observation

Proposal 10752, Observation 5: MIRI-IFU-BKG

Tue Jun 02 18:00:22 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [MIRI-IFU (Obs 3), MIRI-IFU (Obs 4)]

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	SGR-C-BKG	RA: 17 44 43.0000 (266.1791667d) Dec: -29 28 3.00 (-29.46750d) Equinox: J2000		

Comments: Background target for MIRI observations. The exact position will be selected in close conjunction with the STScI team in phase II.
Category=Star
Description=[Ejecta, Young stellar objects]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F560W	FASTR1	30	1	1	Dither 1	4	4	333.005	
1	SHORT(A)	MRSLONG		FASTR1	36	1	1	Dither 1	4	4	399.606	174788
1	SHORT(A)	MRSSHORT		FASTR1	36	1	1	Dither 1	4	4	399.606	
2		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	
2	MEDIUM(B)	MRSLONG		FASTR1	36	1	1	Dither 1	4	4	399.606	
2	MEDIUM(B)	MRSSHORT		FASTR1	36	1	1	Dither 1	4	4	399.606	
3		IMAGER	F1000W	FASTR1	30	1	1	Dither 1	4	4	333.005	
3	LONG(C)	MRSLONG		FASTR1	36	1	1	Dither 1	4	4	399.606	
3	LONG(C)	MRSSHORT		FASTR1	36	1	1	Dither 1	4	4	399.606	

Special Requirements	
	Group Observations 1, 2, 3, 4, 5, Non-interruptible